Mandelbrot Float Renderer Benchmark Report

May 7, 2026

Method

This report benchmarks the CPU renderers with a fixed scene centered at (-0.743643887037151, 0.131825904205330) and zoom 180.0. Each configuration runs 5 times, with 40 warmup frames and 240 measured frames per run. Single-threaded variants are pinned to CPU list `3`; the threaded renderer is pinned to `0-15`. Telemetry is sampled every 0.50 seconds, with a configured cooldown of 0.0 seconds between runs. The benchmark shell requested governor performance before running the measurements.

Representative data requires at least 5 runs, matching checksums, at least 60 measured frames per run, no thermal-throttle counter increments, and a between-run coefficient of variation no higher than 5.0%.

Summary Table

CPU Compute

Opt	Variant	Mean ms	Std ms	CV %	Base %	Prev %	RSS MiB	Rep	Match
-O1	naive	106.900	0.224	0.21	+0.0	–	19.69	yes	yes
-O1	arrays	498.299	12.810	2.57	-366.1	-366.1	19.94	no	no
-O1	simd	100.786	0.603	0.60	+5.7	+79.8	19.94	yes	no
-O1	simd_threads	11.729	0.044	0.37	+89.0	+88.4	19.94	yes	no
-O2	naive	99.707	0.575	0.58	+0.0	–	19.94	yes	yes
-O2	arrays	107.651	0.283	0.26	-8.0	-8.0	19.94	yes	no
-O2	simd	48.462	0.211	0.43	+51.4	+55.0	19.94	yes	no
-O2	simd_threads	10.570	0.034	0.32	+89.4	+78.2	19.94	yes	no
-O3	naive	100.314	1.441	1.44	+0.0	–	19.94	yes	yes
-O3	arrays	56.826	0.137	0.24	+43.4	+43.4	19.94	yes	no
-O3	simd	49.283	0.259	0.53	+50.9	+13.3	19.94	yes	no
-O3	simd_threads	10.633	0.052	0.49	+89.4	+78.4	19.94	yes	no

Hidden-Window Presentation

Variant	n	Render ms	Present ms	Total ms	Present % total
naive	2	230.530	1.671	232.201	0.72
arrays	2	127.401	1.614	129.015	1.25
simd	2	112.570	1.632	114.202	1.43

Variant	n	Render ms	Present ms	Total ms	Present % total
simd_threads	2	10.456	1.630	12.086	13.49

GPU Render

Variant	n	Render ms
gpu	3	7.507

Uncertainty

Repeated-run uncertainty is reported for CPU, hidden-window, and GPU measurements using:

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\mathrm{SEM} = \frac{\sigma}{\sqrt{n}}, \quad \mathrm{CI}_{95} = t_{0.975,n-1} \cdot \frac{\sigma}{\sqrt{n}}, \quad \mathrm{Inacc.\%} = 100 \cdot \frac{\mathrm{CI}_{95}}{\bar{x}}$$

CPU Compute

Opt	Variant	n	Mean ms	$-\sigma$ ms	$+\sigma$ ms	SEM ms	95% ± ms	Inacc. %
-O1	naive	5	106.900	106.675	107.124	0.100	0.279	0.26
-O1	arrays	5	498.299	485.489	511.109	5.729	15.903	3.19
-O1	simd	5	100.786	100.183	101.389	0.270	0.748	0.74
-O1	simd_threads	5	11.729	11.685	11.772	0.019	0.054	0.46
-O2	naive	5	99.707	99.132	100.283	0.257	0.714	0.72
-O2	arrays	5	107.651	107.367	107.934	0.127	0.352	0.33
-O2	simd	5	48.462	48.252	48.673	0.094	0.261	0.54
-O2	simd_threads	5	10.570	10.536	10.604	0.015	0.042	0.40
-O3	naive	5	100.314	98.873	101.755	0.644	1.789	1.78
-O3	arrays	5	56.826	56.689	56.963	0.061	0.170	0.30
-O3	simd	5	49.283	49.024	49.542	0.116	0.322	0.65
-O3	simd_threads	5	10.633	10.581	10.684	0.023	0.064	0.60

Hidden-Window Presentation

Variant	n	Render ms	σ_R	Present ms	σ_P	Total ms	σ_T	95% $\pm_T$	Inacc. %
naive	2	230.530	0.701	1.671	0.058	232.201	0.759	6.816	2.94
arrays	2	127.401	0.077	1.614	0.002	129.015	0.079	0.709	0.55
simd	2	112.570	0.054	1.632	0.022	114.202	0.076	0.686	0.60
simd_threads	2	10.456	0.096	1.630	0.048	12.086	0.144	1.293	10.70

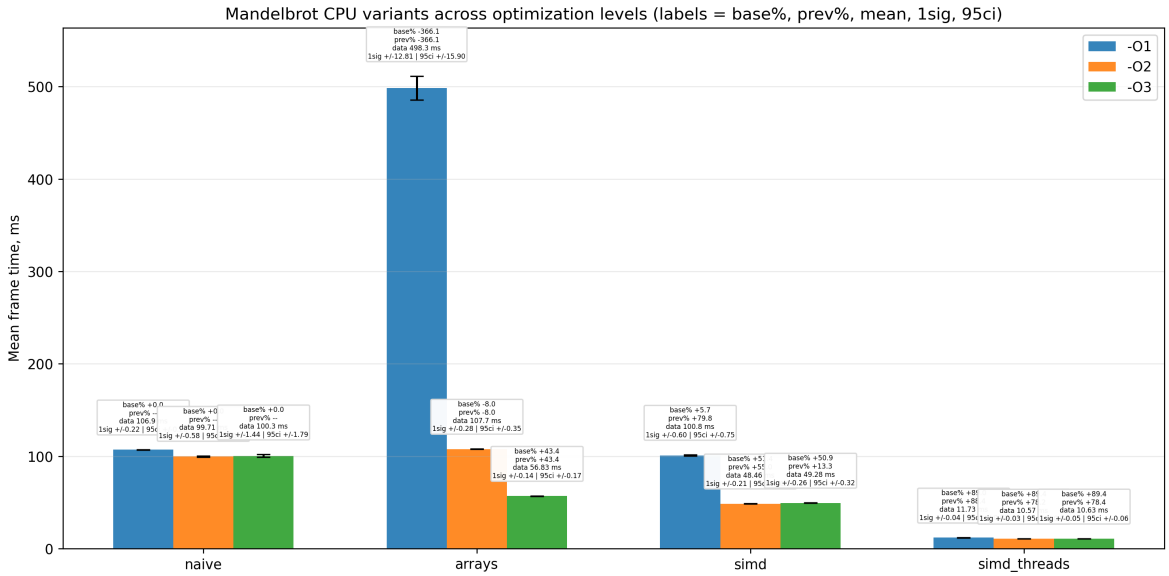
GPU Render

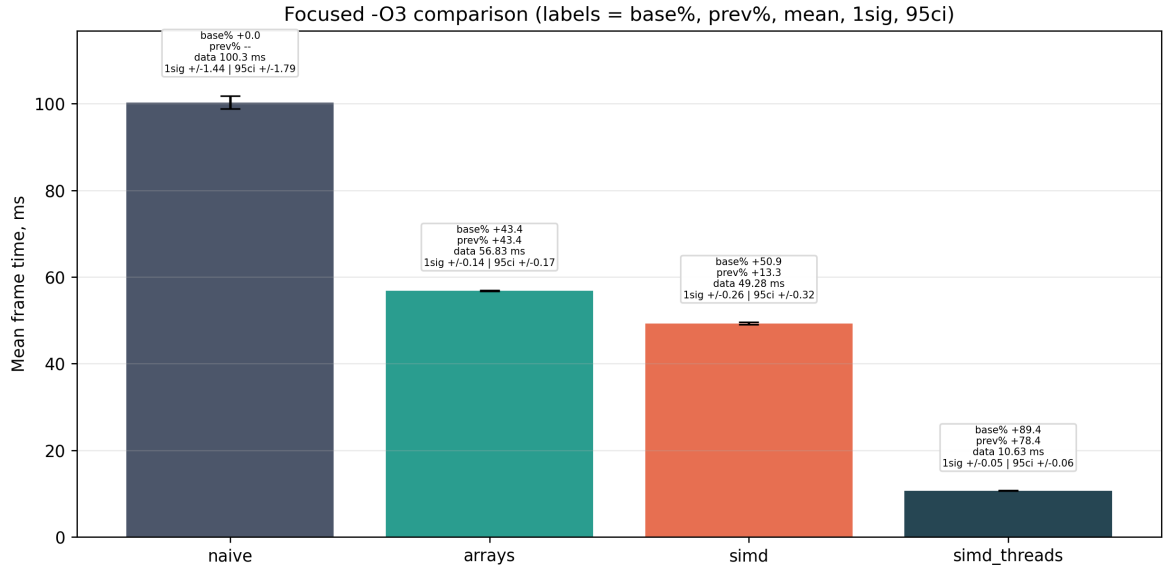
Variant	n	Render ms	σ_R	SEM ms	95% $\pm_R$	Inacc. %
gpu	3	7.507	0.249	0.144	0.619	8.24

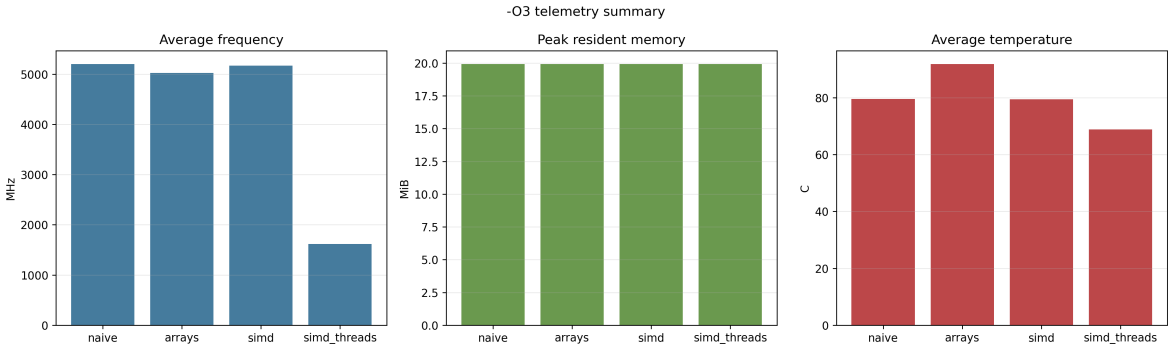
Environment

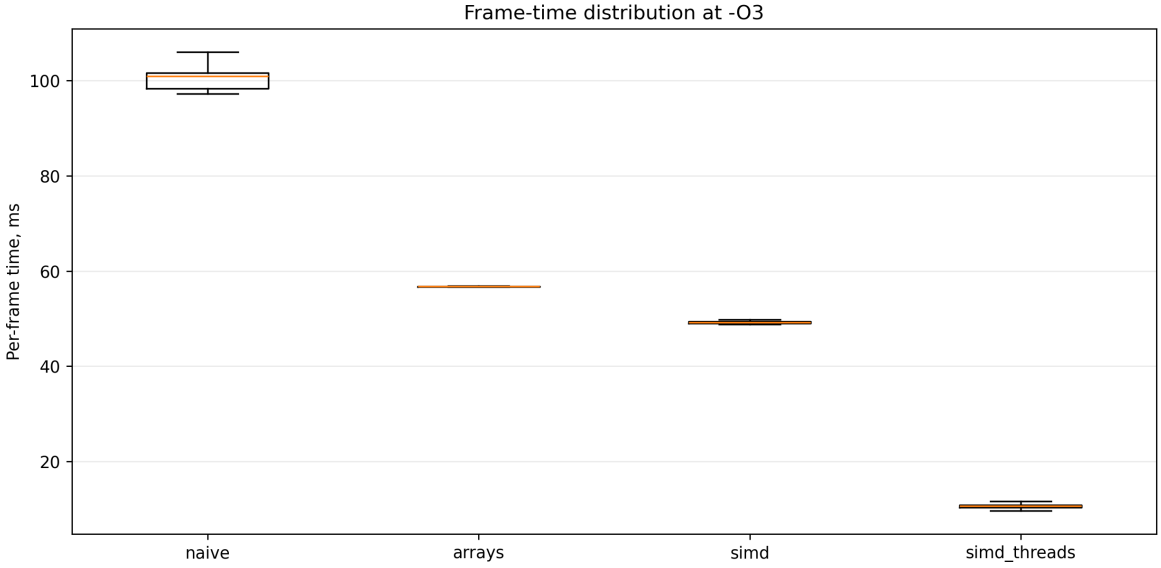
```
analyzing CPU 7:
  current policy: frequency should be within 400 MHz and 4.50 GHz.
                  The governor "performance" may decide which speed to use
                  within this range.

Architecture:                         x86_64
CPU op-mode(s):                       32-bit, 64-bit
Address sizes:                        46 bits physical, 48 bits virtual
Byte Order:                           Little Endian
CPU(s):                               16
On-line CPU(s) list:                  0-15
Vendor ID:                            GenuineIntel
Model name:                           Intel(R) Core(TM) Ultra 9 285H
CPU family:                           6
Model:                                197
Thread(s) per core:                   1
Core(s) per socket:                   16
Socket(s):                            1
Stepping:                             2
CPU(s) scaling MHz:                   69%
CPU max MHz:                          5400,0000
CPU min MHz:                          400,0000
BogoMIPS:                             7372,80
Virtualization:                       VT-x
L1d cache:                            480 KiB (12 instances)
L1i cache:                            768 KiB (12 instances)
L2 cache:                             28 MiB (9 instances)
L3 cache:                             24 MiB (1 instance)
```

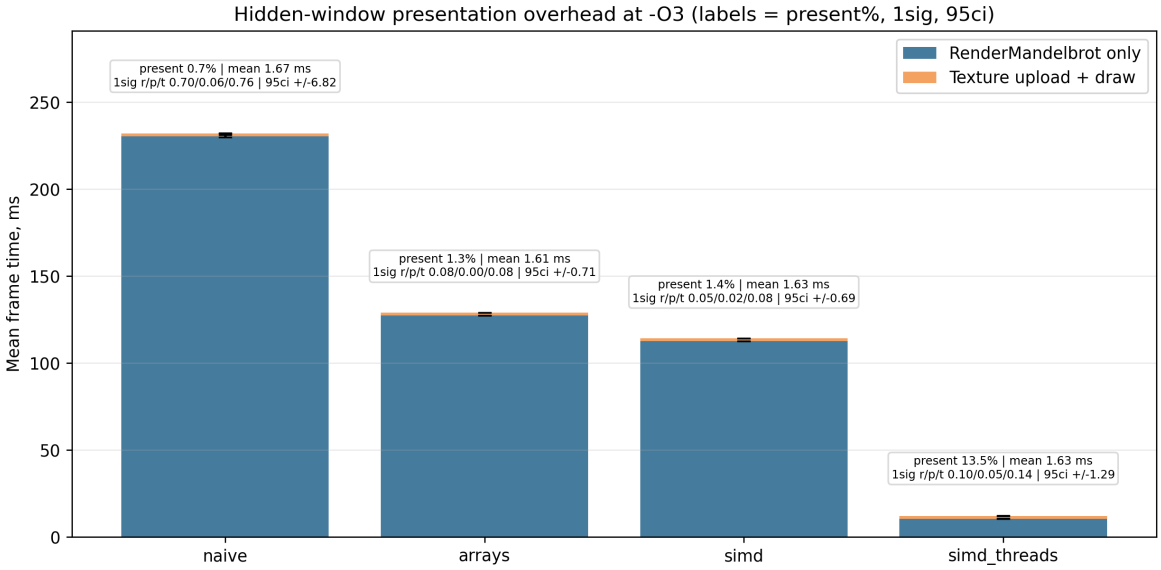
Plots

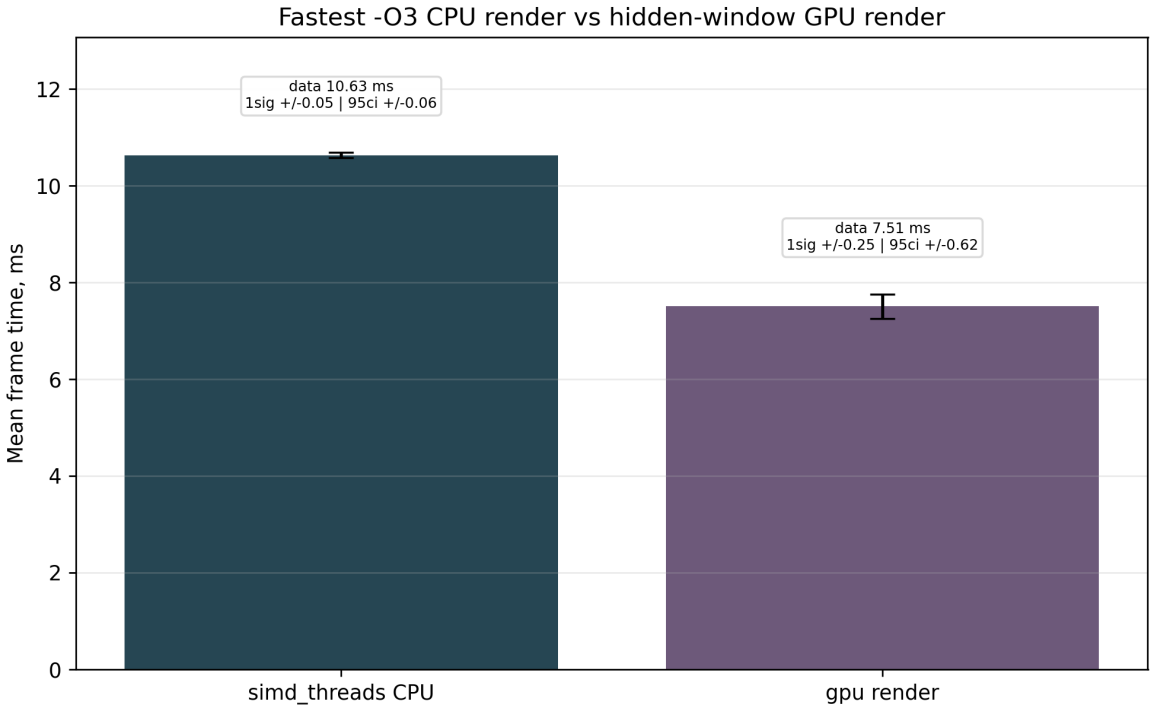

Mandelbrot CPU variants across optimization levels (labels = base%, prev%, mean, 1sig, 95ci)
Mean frame time, ms
-O1
-O2
-O3
base% -366.1
prev% -366.1
data 498.3 ms
1sig +/-12.81 | 95ci +/-15.90
base% -8.0
prev% -8.0
data 107.7 ms
1sig +/-0.28 | 95ci +/-0.35
base% +43.4
prev% +43.4
data 56.83 ms
1sig +/-0.14 | 95ci +/-0.17
base% +0.0
prev% --
data 100.3 ms
1sig +/-1.44 | 95ci +/-1.79
base% +5.7
prev% +79.8
data 100.8 ms
1sig +/-0.60 | 95ci +/-0.75
base% +50.9
prev% +13.3
data 49.28 ms
1sig +/-0.26 | 95ci +/-0.32
base% +89.4
prev% +78.4
data 10.63 ms
1sig +/-0.05 | 95ci +/-0.06
naive
arrays
simd
simd_threads


Focused -O3 comparison (labels = base%, prev%, mean, 1sig, 95ci)
Mean frame time, ms
base% +0.0
prev% --
data 100.3 ms
1sig +/-1.44 | 95ci +/-1.79
base% +43.4
prev% +43.4
data 56.83 ms
1sig +/-0.14 | 95ci +/-0.17
base% +50.9
prev% +13.3
data 49.28 ms
1sig +/-0.26 | 95ci +/-0.32
base% +89.4
prev% +78.4
data 10.63 ms
1sig +/-0.05 | 95ci +/-0.06
naive
arrays
simd
simd_threads


-O3 telemetry summary
Average frequency
MHz
0
1000
2000
3000
4000
5000
naive
arrays
simd
simd_threads
Peak resident memory
MiB
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
naive
arrays
simd
simd_threads
Average temperature
C
0
20
40
60
80
naive
arrays
simd
simd_threads


Frame-time distribution at -O3
Per-frame time, ms
20
40
60
80
100
naive
arrays
simd
simd_threads


Hidden-window presentation overhead at -O3 (labels = present%, 1sig, 95ci)
RenderMandelbrot only
Texture upload + draw
Mean frame time, ms
0
50
100
150
200
250
present 0.7% | mean 1.67 ms
1sig r/p/t 0.70/0.06/0.76 | 95ci +/-6.82
present 1.3% | mean 1.61 ms
1sig r/p/t 0.08/0.00/0.08 | 95ci +/-0.71
present 1.4% | mean 1.63 ms
1sig r/p/t 0.05/0.02/0.08 | 95ci +/-0.69
present 13.5% | mean 1.63 ms
1sig r/p/t 0.10/0.05/0.14 | 95ci +/-1.29
naive
arrays
simd
simd_threads


Fastest -O3 CPU render vs hidden-window GPU render
Mean frame time, ms
0
2
4
6
8
10
12
data 10.63 ms
1sig +/-0.05 | 95ci +/-0.06
data 7.51 ms
1sig +/-0.25 | 95ci +/-0.62
simd_threads CPU
gpu render